

1 Literature Review

1.1 Knowledge Graph-Based Approaches for Fake News Detection

The integration of knowledge graphs (KGs) has emerged as a transformative approach in fake news detection, addressing the limitations of traditional text-based methods that lack factual verification capabilities. Early foundational work by [8] introduced DEAP-FAKED, which combined natural language processing with tensor decomposition to encode both news content and KG entities. Their framework achieved F1-scores of 88% and 78% on two benchmark datasets, demonstrating the potential of KG integration for improving detection accuracy.

Building upon this foundation, recent research has explored more sophisticated KG architectures. [10] proposed KEHGNN-FD, a knowledge graph-enhanced heterogeneous graph neural network that models news, topics, and entities within a unified framework. Their approach outperformed seven baseline methods by effectively aggregating information through attention mechanisms. Similarly, [3] developed KG-MFEND, an efficient knowledge graph-based model for multi-domain fake news detection that constructs comprehensive multi-domain KGs to enhance generalization capabilities across different contexts.

1.2 Attention Mechanisms and Graph Neural Networks

The incorporation of attention mechanisms has significantly advanced the field by enabling models to focus on the most relevant information for fake news detection. [5] developed a recommender system leveraging KGs and graph attention networks that achieved an impressive F1-score of 95%, outperforming classical machine learning and transformer-based models while demonstrating robustness against concept drift over time.

Knowledge-aware attention mechanisms have shown particular promise. [2] introduced KAN (Knowledge-aware Attention Network), which aligns entity mentions in news with KG entities and employs specialized attention modules to weigh the importance of knowledge-level relationships. This approach yielded superior performance across multiple datasets. More recently, [7] proposed KGFakeNet, which combines triplet extraction from statements with attention over GenAI-generated embeddings, achieving improvements of

4% accuracy and 5% F1-score over large language model baselines.

1.3 Multi-Modal and Cross-Domain Approaches

Recent advancements have expanded into multi-modal and cross-domain fake news detection, integrating textual, visual, and social context features with knowledge graphs. [6] introduced FKGFND, which enhances multi-modal fake news detection through fine-grained knowledge graph fusion. Concurrently, [1] developed SCCN, a multimodal semantic-enhanced attention network that utilizes co-attention mechanisms to fuse information from multiple sources, significantly boosting detection accuracy and generalization across domains.

The challenge of cross-domain generalization has been addressed through innovative architectures. [9] proposed a multi-knowledge and LLM-inspired heterogeneous graph neural network that leverages large language models for feature extraction, outperforming state-of-the-art baselines in diverse settings. These approaches demonstrate the evolving sophistication of KG-based methods in handling the complex, multi-faceted nature of fake news across different domains and modalities.

1.4 Interpretability and Commonsense Knowledge Integration

A significant trend in recent literature emphasizes model interpretability through commonsense knowledge integration. [4] developed an interpretable fake news detection method based on commonsense knowledge graphs that not only achieves accuracy up to 91% but also provides transparent reasoning by highlighting which external facts support or contradict news content. This focus on explainability addresses the critical need for trustworthy fake news detection systems that can provide justification for their classifications.

The integration of commonsense knowledge with graph matching techniques has proven particularly effective for creating models that are both accurate and interpretable. These approaches enable the identification of logical inconsistencies between news content and established commonsense knowledge, providing a robust mechanism for fake news detection that aligns with human reasoning processes.

1.5 Research Gaps and Future Directions

Despite significant advancements, several research gaps remain. First, while recent approaches show improved performance, many still require extensive computational resources and large training datasets, limiting their practical deployment [8]. Second, most existing methods use simplistic knowledge integration techniques such as concatenation, failing to dynamically weigh the relevance of different knowledge elements to specific contexts. Third, the generalization of models across diverse cultural and linguistic contexts remains

challenging due to variations in knowledge representation and commonsense understanding.

Our proposed KNOW-NET architecture directly addresses these gaps through computational efficiency achieved via cross-attention mechanisms combined with pre-trained language models, enabling state-of-the-art performance with minimal training data and rapid convergence. Additionally, KNOW-NET implements intelligent knowledge integration through dynamic attention weighting that allows the model to focus on the most relevant KG entities for each specific news context, moving beyond static concatenation approaches used in prior work. The multi-task learning framework further enhances feature learning and model generalization through joint optimization of fake news classification and named entity recognition.

Future research should focus on developing more efficient KG construction and updating mechanisms, exploring cross-lingual and cross-cultural knowledge integration, and enhancing model robustness against deliberately crafted misinformation. Additionally, the integration of temporal dynamics in knowledge graphs to account for evolving facts and contexts represents a promising direction for addressing the challenge of "temporal fake news" where information may be contextually accurate but temporally misaligned.

References

- [1] W. Chen, Y. Dang, and X. Zhang. A multimodal semantic-enhanced attention network for fake news detection. *Entropy*, 27(7):746, 2025.
- [2] Y. Dun, K. Tu, C. Chen, C. Hou, and X. Yuan. Kan: Knowledge-aware attention network for fake news detection. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 35, pages 81–89, 2021.
- [3] L. Fu, H. Peng, and S. Liu. Kg-mfend: an efficient knowledge graph-based model for multi-domain fake news detection. *The Journal of Supercomputing*, 79(15):1–28, 2023.
- [4] X. Gao, W. Chen, L. Lu, Y. Cui, X. Dai, L. Dai, K. Wang, J. Shen, Y. Wang, S. Wang, Z. Yu, and H. Liu. An interpretable fake news detection method based on commonsense knowledge graph. *Applied Sciences*, 13(11):6680, 2023.
- [5] A. Golovin, N. Zhukova, R. Delhibabu, and A. Subbotin. Improving recommender systems for fake news detection in social networks with knowledge graphs and graph attention networks. *Mathematics*, 13(6):1011, 2025.
- [6] R. Hao, H. Luo, and Y. Li. Multi-modal fake news detection enhanced by fine-grained knowledge graph. *IEICE Transactions on Information and Systems*, 108(4):604–614, 2025.

- [7] A. Kumar, P. Kumar, A. Yadav, S. Ahlawat, and Y. Prasad. Kgfakenet: A knowledge graph-enhanced model for fake news detection. *Journal of Artificial Intelligence Research*, 75:109–122, 2025.
- [8] Mohit Mayank, Shakshi Sharma, and Rajesh Sharma. Deap-faked: Knowledge graph based approach for fake news detection. In *2022 IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining (ASONAM)*, pages 47–51, 2021.
- [9] B. Xie, X. Ma, X. Shan, A. Beheshti, J. Yang, H. Fan, and J. Wu. Multiknowledge and llm-inspired heterogeneous graph neural network for fake news detection. *IEEE Transactions on Computational Social Systems*, 12(3):682–694, 2025.
- [10] B. Xie, X. Ma, J. Wu, J. Yang, and H. Fan. Knowledge graph enhanced heterogeneous graph neural network for fake news detection. *IEEE Transactions on Consumer Electronics*, 70(12):2826–2837, 2024.